Exp No: 25

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PRODUCER–CONSUMER PROBLEM USING SYSTEMC

AIM:

To implement and simulate the Producer–Consumer problem using SystemC.

SOFTWARE REQUIRED:

Ubuntu / WSL Linux g++

Compiler CMake **SystemC**

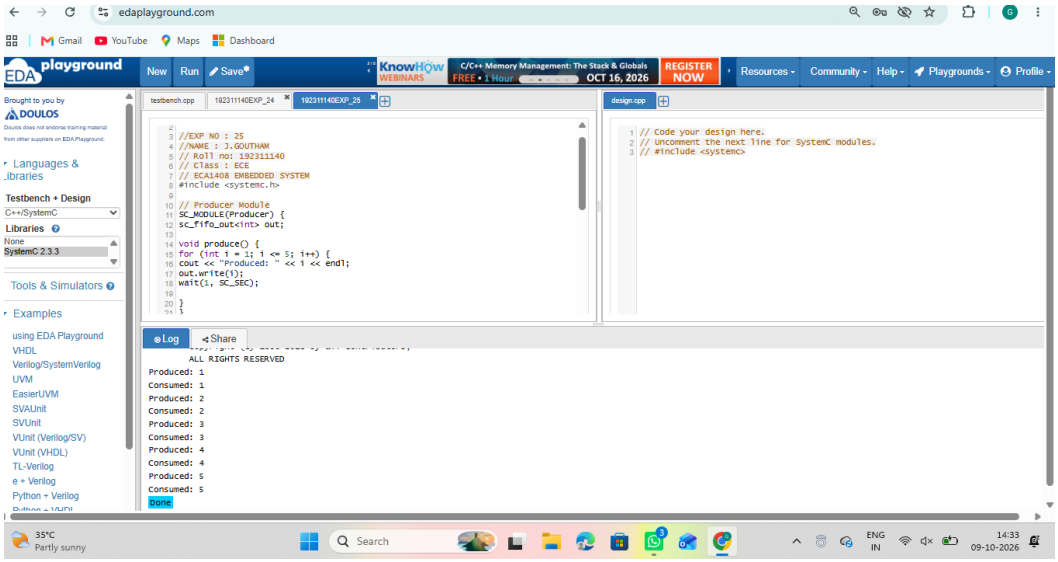
Library

PROGRAM:

```
#include <systemc.h>
// Producer Module SC_MODULE(Producer)
{
sc_fifo_out<int> out; void produce()
{ for (int i = 1; i <= 5; i++) { cout <<
"Produced: " << i << endl;
out.write(i);
wait(1, SC_SEC);
}
}
SC_CTOR(Producer) {
SC_THREAD(produce);
}
};
// Consumer Module
SC_MODULE(Consumer) {
```

```
sc_fifo_in<int> in;
void consume() {
int value; while
(true) {
in.read(value);
cout << "Consumed: " << value << endl;
wait(1, SC_SEC);
}
}
SC_CTOR(Consumer) {
SC_THREAD(consume);
}
};
int sc_main(int argc, char* argv[]) {
sc_fifo<int> fifo(5);
Producer producer("Producer"); Consumer
consumer("Consumer");
producer.out(fifo);
consumer.in(fifo); sc_start(10,
SC_SEC); return 0;
}
```

OUTPUT:



RESULT:

Thus, the Producer–Consumer problem was successfully implemented and simulated using SystemC. Proper synchronization between concurrent processes was achieved using FIFO communication.